

Possible Method to get k1,2,3 keywords in kryptos using the extra E's/dots in the k0 morse code

I have found a method to get the k1 and k2 keywords directly from the k0 morse sections using the extra dots or 'e's around the morse code words. The approach might also give the matrix sizes for k3. It is quite simple such that I initially thought it could not be new however after searching for quite a while I cannot find this method written down anywhere. It would be great to get this method cross checked for errors and to know if it has previously been documented hence posting it around to get some feedback.

Method to get k1 keyword from k0

Note I am referring here to words as top / bottom for each morse code part. E.g. in "Virtually Invisible" the top word is "virtually" and the bottom word is "invisible".

1. count the extra dots either side of each word in k0 and get a number for that word. If there are dots on both sides of the word add them together.
2. Use that number to count a letter inside that word.
 - a. for the top word count from the end / right of the word.
 - b. for the bottom word count from the start / left of the word.
3. If there are no dots then take the letter to the left of the very middle letter of the word or phrase (put all the words together for "tisyour")
4. Ignore words of length 2 or 3 (i.e. RQ and SOS)

Below is each word with the number next to each word being the number of E's/dots on the morse code (before and after) . Then the total if the dots are before and after, and the "letter" that this number selects from the word.

2 VIRTUALLY 1 / 2+1 = 3 / 3 from right is "L"
6 INVISIBLE / 6 from the left is "I"
2 SHADOW 2 / 2 + 2 = 4 / 4 from the right is "A"
FORCES 5 / 5 from the left is "E"
LUCID 3 / 3 from the right is "C"
MEMORY 1 / 1 from the left is "M"
DIGITAL 3 / 3 from the right is "T"
INTERPRETATI / no dots / the left of middle of the word (between P and R) is "P"
T IS YOUR / no dots / left of the middle letter of the phrase (Y) is "S"
POSITION 1 / 1 from the left is "P"

so we get L I A E C M T P S P which anagrams to PALIMPCEST the k1 keyword albeit the "incorrect" keyword we find at the end used to encode Iqlusion.

When you combine this with the photos of the k1 crib sheet that Jim revealed I think he might be telling a story about how the "cryptographer" initially encoded the k1 PT with the "PALIMPCEST" keyword (with a spelling mistake) , then made their "stego" code in k0 to record that keyword. Then they realised their spelling mistake and erased all the "Cs" and made them "Ss" and redid the k1 encoding, except they missed the very last C and they didn't fix the stego in k0. You can even stretch to think of this

as an old code “peeking through” from underneath the new code – like a “Palimpsest”

There are two questions about the exact morse transcription that would be great to check and confirm.

- i. Is there a dot before DIGETAL ? Online I have found both yes and no. Zooming into the photo there is some sort of mark there but is it really an “E”, or is it a bolt or screw head or other kind of mark ? For the method above to work there has to be no dot there. DIGETAL needs 3 dots to get the T counting from the right and there are clearly 3 dots after the end of the word.
- ii. Is there a dot at the end of POSITION ? Online most places say yes although it is very hard to see in the photos. However one of the online pages says the E at the end is trimmed so it is often overlooked which seems pretty definitive. For this method we need the dot to be there to get the letter P one from the front.

I have to say also I am not in love with rule 3. It’s not that intuitive in the way the other rules seem at least to me.

Method to get k2 keyword from k1

If you write down k1 – but using rule 4 to ignore all the 2 and 3 letter words – you get a sentence with just 8 words. Take the 8 numbers found from k0 - 3,6,4,5,3,1,3,1 – And then using these we extract a letter from each of these 8 words in k1 counting from the start of the word.

Between – 1 - B
Subtle – 1 - S
Shading – 3 - A
Absence - 6 - C
Light – 3 - G
Lies – 4 - S
Nuance - 3 - A
Iqlusion – 5 - S

You get ABSCGSSA. However the word LIGHT giving the G would give I if we change the 3 to a 2. So 1,1,3,6,2,4,3,5

Between – 1 - B
Subtle – 1 - S
Shading – 3 - A
Absence - 6 - C
Light – 2 - I
Lies – 4 - S
Nuance - 3 - A
Iqlusion – 5 - S

And you get “Abscissa” the keyword for k2. However we had to make that change which is suspect and I could not find a way to give the order for these numbers to be applied to the words so it could quite likely just be an accidental pattern found knowing what the answer is.

Method to get k3 matrix sizes from k2

Finally is a way to find the “keys” to k3 from k2. In the case of k3 the keys to the transposition would be 8 and 24 being the matrix size Jim used for the two rotations.

Given we want to count characters following our methods above, and also that the keyword is “abscissa” (meaning the “x-axis” value of point on a graph) you can count the horizontal (x-axis) position of the 4 possible “markers” in k2 – one is the “mistake” character U in “underground” plus the three “?” in k2.

1. The “U” mistake is the “R” in ..QCRTB.. and is 24 letters from the left and 9 from the right. (does the “R” mean “Rotate”?)
2. The first ? is 7 from the left and 24 from the right
3. The second ? is 9 from the left and 23 from the right
4. The third ? is 8 from the left and 24 from the right

It could be the third ? give the rotation keys to k3 of 8 and 24. Its quite nice how the “from the left and from the right” idea used in k0 comes back again. Or maybe it is the “R” (24/9) but he made another small “mistake” ? If that line was one character shorter it would work. The “R” would also be nice given the comment Jim made “... it’s the position or orientation of these mistakes that is important” ? Either way the main appeal of a method like this is that it aligns conceptually with “counting characters left and right”. It is however less clear here if any of this is intentional.

Possible Implications for k4 ?

All this suggests that counting a particular number of letters into words or phrases to get a character could be important for solving k4. Maybe there is a long non-english key (Vigenère or OTP) for k4 that could be extracted from the k321 text using these sorts of methods ? Is the number sequence 1,1,3,6,2,4,3,5. (or 1,1,3,6,3,4,3,5) relevant to help pick out particular letters to make a key. ? The position of the 4th ? in the sculpture – is 27 from the left and 5 from the right – does this suggest a transposition being involved somewhere in k4 ? I even get the sense “making little mistakes that have some further meaning” is part of story being told in kryptos so there might be a small mistake to correct in getting to solve k4.

There are a few comments I’ve read from Jim (or Ed?) that seem relevant to all this. In particular. “k1 and k2 were not solved how I intended” and “in an analog system like kryptos the algorithm is the key”. I take both of these to imply that there are some rules that need to be discovered in order to tackle k4. There was also a comment about “clues that are hidden in plain sight” which appears relevant if in particular the k0 to k1 method above turns out to be intentionally there.

I did pay and check with Jim the k_0 to k_1 part with Jim but quite reasonably got the response ... "i am not in the habit of commenting on methods, that is giving clues, i have given all the clues i will give."

If you have made it this far then thanks heaps for reading all the way through. I would love to hear any feedback positive or negative. If there mistakes in the initial transcription, or you see holes or errors in the method or arguments please let me know.

Cheers
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